

ULIP Integration Requirement Document

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Project Name: ULIP

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Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	03/02/2026	Sudhanshu Singh	Nayab Ali	Baseline Version

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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for PESO system information from ULIP system.

This document details the following points:

1. **Data Integration between PESO and ULIP web service:** Identification of operational points where data integration between **PESO** and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

Users will share information such as **Vehicle Registration Number** based on data sent by user ULIP system will call respective **PESO** API for accessing information.

Following is the point for accessing vehicle information: -

1. This API will share data based on information provided by users.
2. ULIP will expose API for getting data from PESO API.
3. **PESO/01** API will take Vehicle Registration Number from user and get data from PESO API.

1.2 Data Integration between PESO System and ULIP application

Users will share information with the ULIP system based on data shared ULIP will provide data to user.

1.3 PESO/01

This API will take **Vehicle Registration Number** and connect with PESO API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	Vehicle Registration Number	Enter the Vehicle Registration Number as available in PESO system.	^[A-Z0-9]{5,11}\$	Required	8

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide Vehicle Registration Number for which detail is required.
3. PESO API will share details of CNG certificate details of vehicles based on Vehicle Registration Number which exist in PESO system in JSON format.
4. PESO Integration will add JSON in **response** key in response body.
5. ULIP system will provide the data in JSON Object format in response body.

Example: Vehicle Registration Number is shared by ULIP system, then PESO-integration will make a request to PESO API.

Request: The request must be of the following format, where Vehicle Registration Number is passed within the URL body in JSON format.

For Staging: <https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/PESO/01>

```
{ " VehicleRegistrationNumber": " HR38AC7120"}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/PESO/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJhcHBzIjoieXBpR2F0ZXdeSIsInN1Yil6InVsaXAiLCJpYXQiOiJlE3Nz
AwOTE4NTd9.3ku5gqG3yAwOazqC5EA69PEiiZmDymndimvRfr1ICz6nFjvLHbRw2LrCCUz2e
dQmZt_dZj0B8x-jrth157OQ' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{"VehicleRegistrationNumber": "HR38AC7120"}'
```

For Production: <https://www.ulip.dpiit.gov.in/ulip/v1.0.0/PESO/01>

```
{" VehicleRegistrationNumber": " HR38AC7120"}
```

Curl URL

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/PESO/01' \
--header 'authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJhcHBzIjoieXBpR2F0ZXdeSIsInN1Yil6InVsaXAiLCJpYXQiOiJlE3Nz
AwNDE2NDV9.iERfbuYgZbDbeuVOOUB89TfO3ttZnnaTB8_kSGj3R12gID8IVz0cLVL2EvADO
mEu1hZtbn8P4ivymEJCdG9FAA' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data '{"VehicleRegistrationNumber": "HR38AC7120"}'
```

For obtaining authentication token, refer [Section 1.4](#)

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Response

To acknowledge the above request, ULIP will send the response below -

1. Invalid Vehicle Registration Number

```
{  "response": null,
  "error": "true",
  "code": "400 BAD_REQUEST",
  "message": "Data format failed OR wrong value entered at: VehicleRegistrationNumber.
Format should follow ^[A-Z0-9]{5,11}$" }
```

2. In case Vehicle Registration Number exists

```
{  "response": [
    {
      "response": [
        {
          "ErrorMsg": "",
          "Cylinder Manufacturing Date": "01/10/2021",
          "Cylinder Make": "M/s. Zhejiang Jindun Pressure Vessel Co. Ltd., China",
          "Cylinder capacity in water litre": 150.0,
          "Number of cylinders": "Not Available at PESO",
          "Cylinder ID/ Serial Number": "K-6834833",
          "Hydro Test Date": "26/12/2025",
          "Hydro Test Due Date": "25/12/2028",
          "Test Result": "Pass",
          "Cylinder Age valid till Date": "Not Available at PESO",
          "Testing Company Name": "MAHADEV CNG CYLINDER TESTING COMPANY",
          "Certificate No": "G145658/2025/2579",
          "Vehicle Registration Number": "HR38AC7120"
        },
        {
          "ErrorMsg": "",
          "Cylinder Manufacturing Date": "01/10/2021",
          "Cylinder Make": "M/s. Zhejiang Jindun Pressure Vessel Co. Ltd., China",
          "Cylinder capacity in water litre": 150.0,
          "Number of cylinders": "Not Available at PESO",
          "Cylinder ID/ Serial Number": "K-6834166",
          "Hydro Test Date": "26/12/2025",
          "Hydro Test Due Date": "25/12/2028",
          "Test Result": "Pass",
          "Cylinder Age valid till Date": "Not Available at PESO",
          "Testing Company Name": "MAHADEV CNG CYLINDER TESTING COMPANY",
          "Certificate No": "G145658/2025/2580",
          "Vehicle Registration Number": "HR38AC7120"
        }
      ]
    }
  ],
}
```

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```
"responseStatus": "SUCCESS"
},
"error": "false",
"code": "200",
"message": "Success"}
```

3. In case VehicleRegistrationNumber not exist

```
{ "response": [
  {
    "response": [
      {
        "ErrorMsg": "No Record found"
      }
    ],
    "responseStatus": "SUCCESS"
  }
],
"error": "false",
"code": "200",
"message": "Success"}
```

Status Code 200: PESO data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server errors occurred

Status Code 502: Server is not responding.

1.4 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in PESO system through a data exchange URL that has following form:

For Staging: <https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/PESO/01>

```
{" VehicleRegistrationNumber": " HR38AC7120"}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until tokens will not get expired.

Summary of HTTP Basic Authentication is as follows:

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First time User hits ULIP secured API through username and password for access token as follows-

```
curl --location --request POST
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "username": "xxxx",
    "password": "xxxx@123"}'
```

Authorization:

Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNjgyNDkyNzQ1LCJhcHBzljoiZGF0YXB1c2gifQ.iNjO9ggx99yqkjJw65KKP7dZ9zVatpuQQP76rVBnxxArEYq8QZIn48_ihG8ECe34wKm3QTcokX7wrq9RDwZrbA'

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represents user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, users can request ULIP system for cng certificate details as follow -

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/PESO/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJhcHBzljoiYXBpR2F0ZXdhcSIsInN1Yil6InVsaXAiLCJpYXQiOiJlE3NDI3NDY0ODV9.vYb9i-
tHSUe0RXjO2ETOywsBrhPQ4vN1f77HnqAODcLtsVlcUnLNcOEU25ZdVc15UI360Eh4NgIA
wkwwui7BFA' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{" VehicleRegistrationNumber": " HR38AC7120"}'
```

For Production:

```
curl --location --request POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \
  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "username": "xxxx",
    "password": "xxxx@123"}'
```

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Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWUiOiJ1bGlwliwiaWF0IjoxNjgyNDkyNzQ1LCJhcHBzIjoieZGF0YXB1c2gifQ.iNjO9ggx99yqkJw65KKP7dZ9zVatpuQQP76rVBnxxArEYq8QZIn48_ihG8ECe34wKm3QTcokX7wrq9RDwZrbA'

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```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/PESO/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJhcHBzIjoieZXBpR2F0ZXdheSIsInN1Yil6InVsaXAiLCJpYXQiOiJE  
3NDI4NzU0MDN9.CknivakcbIjKOUiGga6i1RHUX_nbL174BCrAlihZa6v8phn29lszkNRcXtLK  
dN2sSdkDHS3tqw2-aE0I9-r8XQ' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data '{" VehicleRegistrationNumber": " HR38AC7120"}'
```